MORCHED AMMAR

C++ Python Junior Developer | Embedded Systems | morcheda62@gmail.com | +21623875468 | Ariana, Tunisia
Linkedin@Morched_Ammar | Github@Morched_Ammar

PROFESSIONAL SUMMARY

Ambitious C++ Python Junior Developer with 2+ years of experience in designing, developing, and deploying scalable embedded systems, leveraging C++, Python, and related tools to drive innovation. Proficient in utilizing Python for data analysis, machine learning, and IoT applications, with a strong foundation in C++ for building high-performance embedded systems. Skilled in applying Agile methodologies, version control systems like Git, and collaborative platforms to ensure seamless project execution. Adept at integrating C++ and Python with various libraries, frameworks, and protocols, including OpenCV, ONNX, and MQTT, to create efficient and reliable embedded systems. Passionate about exploring the intersection of C++, Python, and embedded systems to develop cutting-edge solutions.

TECHNICAL SKILLS

C++, Python, OpenCV, ONNX, MQTT, Wi-Fi, I2C, SPI, UART, Git, Agile, Machine Learning, IoT, Embedded Systems, Computer Vision, Data Analytics

PROFESSIONAL EXPERIENCE

Capgemini- Final year intern: mobile Dev Feb 2025/Aout 2024

Capgemini

- Utilized C++ to optimize mobile application performance, resulting in a 25% reduction in latency and a 30% increase in overall efficiency.
- Developed and deployed a Python-based data analytics pipeline, processing over 10,000 data points per second, to inform business decisions and drive growth.
- Collaborated with cross-functional teams to design and implement a scalable IoT architecture, integrating C++ and Python with MQTT and Wi-Fi protocols, to enable real-time data exchange and processing.
- Applied machine learning algorithms using Python and OpenCV to develop a computer vision module, achieving a 95% accuracy rate in object detection and classification.

Sofrecom-Summer internship: embedded system July 2024/ August 2024

Sofrecom

- Designed and developed a C++-based embedded system, leveraging the STM32 microcontroller, to control and monitor industrial equipment, resulting in a 40% reduction in energy consumption and a 25% increase in overall system reliability.
- Implemented a Python-based data visualization module, using ONNX and OpenCV, to provide real-time insights into system performance and enable data-driven decision-making.
- Integrated C++ and Python with various sensors and actuators, including I2C, SPI, and UART, to develop a comprehensive IoT solution, processing over 500 data points per second.
- Utilized Agile methodologies and version control systems like Git to ensure collaborative and efficient project execution, resulting in a 30% reduction in development time and a 25% increase in overall project quality.

PROJECTS

TrafficOS — Deep RL Intersection Controller

Technologies: OpenAI Gymnasium, PPO, Stable Baselines3, Raspberry Pi 5, ONNX

Engineered a custom OpenAI Gymnasium environment modeling a 4-way intersection with an 8-dimensional observation space (per-lane vehicle density + emergency vehicle priority flags). Trained a PPO agent (Stable Baselines3) over 500,000 timesteps, achieving a 30–35% reduction in average vehicle wait time vs. fixed-timer controllers.

Padel Player Tracking — Computer Vision & IoT

Technologies: STM32F407VG, I2C, SPI, MQTT, Wi-Fi

Engineered modular embedded firmware in C on STM32F407VG managing 3 sensors (LEM LTS 6-NP, DHT11, PIR) across I2C and SPI buses with a layered driver architecture enabling sensor hot-swap without core firmware changes. Implemented MQTT-over-Wi-Fi cloud pipeline for continuous telemetry upload, enabling remote monitoring and real-time actuation commands from a cloud dashboard.

IoT Energy Monitoring System — STM32F407VG

Technologies: STM32F407VG, I2C, SPI, MQTT, Wi-Fi

Engineered modular embedded firmware in C on STM32F407VG managing 3 sensors (LEM LTS 6-NP, DHT11, PIR) across I2C and SPI buses with a layered driver architecture enabling sensor hot-swap without core firmware changes. Implemented MQTT-over-Wi-Fi cloud pipeline for continuous telemetry upload, enabling remote monitoring and real-time actuation commands from a cloud dashboard.

EDUCATION

Bachelor's degree in electronics and automation — Higher Institute of Industrial Management — (2018-2022)
Bachelor's degree in computer engineering — Private higher school of engineering and technology — (Sept 2022 - Oct 2025)

CERTIFICATIONS

- Nvidia: Applications of AI for Anomaly Detection (Nov2024)
- Nvidia: Applications of AI for Predictive Maintenance (Nov2024)

LANGUAGES

English (B2), French (B2)